

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)(i)	$(x-5)^2 + (y+2)^2 = \dots$	M1	1.1b
	$(5, -2)$	A1	1.1b
(ii)	$r = \sqrt{5^2 + 2^2 - 11}$	M1	1.1b
	$r = 3\sqrt{2}$	A1	1.1b
	(4)		
(b)	$y = 3x + k \Rightarrow x^2 + (3x+k)^2 - 10x + 4(3x+k) + 11 = 0$ $\Rightarrow x^2 + 9x^2 + 6kx + k^2 - 10x + 12x + 4k + 11 = 0$	M1	2.1
	$\Rightarrow 10x^2 + (6k+2)x + k^2 + 4k + 11 = 0$	A1	1.1b
	$b^2 - 4ac = 0 \Rightarrow (6k+2)^2 - 4 \times 10 \times (k^2 + 4k + 11) = 0$	M1	3.1a
	$\Rightarrow 4k^2 + 136k + 436 = 0 \Rightarrow k = \dots$	M1	1.1b
	$k = -17 \pm 6\sqrt{5}$	A1	2.2a
	(5)		
(9 marks)			
Notes			

(a)(i)

M1: Attempts to complete the square on by halving both x and y terms.

Award for sight of $(x \pm 5)^2, (y \pm 2)^2 = \dots$ This mark can be implied by a centre of $(\pm 5, \pm 2)$.

A1: Correct coordinates. (Allow $x = 5, y = -2$)

(a)(ii)

M1: Correct strategy for the radius or radius². For example award for $r = \sqrt{5^2 + 2^2 - 11}$

or an attempt such as $(x-a)^2 - a^2 + (y-b)^2 - b^2 + 11 = 0 \Rightarrow (x-a)^2 + (y-b)^2 = k \Rightarrow r^2 = k$

A1: $r = 3\sqrt{2}$. Do not accept for the A1 either $r = \pm 3\sqrt{2}$ or $\sqrt{18}$

The A1 can be awarded following sign slips on $(5, -2)$ so following $r^2 = 5^2 + 2^2 - 11$

(b) Main method seen

M1: Substitutes $y = 3x + k$ into the given equation (or their factorised version) and makes progress by attempting to expand the brackets. Condone lack of $= 0$

A1: Correct 3 term quadratic equation.

The terms must be collected but this can be implied by correct a, b and c

M1: Recognises the requirement to use $b^2 - 4ac = 0$ (or equivalent) where both b and c are expressions in k . It is dependent upon having attempted to substitute $y = 3x + k$ into the given equation

M1: Solves 3TQ in k . See General Principles.

The 3TQ in k must have been found as a result of attempt at $b^2 - 4ac \dots 0$

A1: Correct simplified values

Look carefully at the method used. It is possible to attempt this using gradients

(b) Alt 1	$x^2 + y^2 - 10x + 4y + 11 = 0 \Rightarrow 2x + 2y \frac{dy}{dx} - 10 + 4 \frac{dy}{dx} = 0$	M1 A1	2.1 1.1b
	Sets $\frac{dy}{dx} = 3 \Rightarrow x + 3y + 1 = 0$ and combines with equation for C $\Rightarrow 5x^2 - 50x + 44 = 0$ or $5y^2 + 20y + 11 = 0$ $\Rightarrow x = \dots$ or $y = \dots$	M1	3.1a
	$x = \frac{25 \pm 9\sqrt{5}}{5}, y = \frac{-10 \pm 3\sqrt{5}}{5}, k = y - 3x \Rightarrow k = \dots$	M1	1.1b
	$k = -17 \pm 6\sqrt{5}$	A1	2.2a

M1: Differentiates implicitly condoning slips but must have two $\frac{dy}{dx}$'s coming from correct terms

A1: Correct differentiation.

M1: Sets $\frac{dy}{dx} = 3$, makes y or x the subject, substitutes back into C and attempts to solve the resulting quadratic in x or y .

M1: Uses at least one pair of coordinates and l to find at least one value for k . It is dependent upon having attempted both M's

A1: Correct simplified values

(b) Alt 2	$x^2 + y^2 - 10x + 4y + 11 = 0 \Rightarrow 2x + 2y \frac{dy}{dx} - 10 + 4 \frac{dy}{dx} = 0$	M1 A1	2.1 1.1b
	Sets $\frac{dy}{dx} = 3 \Rightarrow x + 3y + 1 = 0$ and combines with equation for l $y = 3x + k, x + 3y = 1$ $\Rightarrow x = \dots$ and $y = \dots$ in terms of k	M1	3.1a
	$x = \frac{-3k-1}{10}, y = \frac{k-3}{10}, x^2 + y^2 - 10x + 4y + 11 = 0 \Rightarrow k = \dots$	M1	1.1b
	$k = -17 \pm 6\sqrt{5}$	A1	2.2a

Very similar except it uses equation for l instead of C in mark 3

M1 A1: Correct differentiation (See alt 1)

M1: Sets $\frac{dy}{dx} = 3$, makes y or x the subject, substitutes back into l to obtain x and y in terms of k

M1: Substitutes for x and y into C and solves resulting 3TQ in k

A1: Correct simplified values

(b) Alt 3	$y = 3x + k \Rightarrow m = 3 \Rightarrow m_r = -\frac{1}{3}$	M1	
	$y + 2 = -\frac{1}{3}(x - 5)$	A1	
	$(x - 5)^2 + (y + 2)^2 = 18, y + 2 = -\frac{1}{3}(x - 5)$ $\Rightarrow \frac{10}{9}(x - 5)^2 = 18 \Rightarrow x = \dots$ or $\Rightarrow 10(y + 2)^2 = 18 \Rightarrow y = \dots$	M1	
	$x = \frac{25 \pm 9\sqrt{5}}{5}, y = \frac{-10 \pm 3\sqrt{5}}{5}, k = y - 3x \Rightarrow k = \dots$	M1	
	$k = -17 \pm 6\sqrt{5}$	A1	

M1: Applies negative reciprocal rule to obtain gradient of radius

A1: Correct equation of radial line passing through the centre of C

M1: Solves simultaneously to find x or y

Alternatively solves " $y = -\frac{1}{3}x - \frac{1}{3}$ " and $y = 3x + k$ to get x in terms of k which they substitute in

$x^2 + (3x + k)^2 - 10x + 4(3x + k) + 11 = 0$ to form an equation in k .

M1: Applies $k = y - 3x$ with at least one pair of values to find k

A1: Correct simplified values

Q2.

Question	Scheme	Marks	AOs
(a)	(i) $x^2 + y^2 - 10x + 16y = 80 \Rightarrow (x-5)^2 + (y+8)^2 = \dots$	M1	1.1b
	Centre $(5, -8)$	A1	1.1b
	(ii) Radius 13	A1	1.1b
		(3)	
(b)	Attempts $\sqrt{5^2 + 8^2} + 13$	M1	3.1a
	$13 + \sqrt{89}$ but fit on their centre and radius	A1ft	1.1b
		(2)	
(5 marks)			
Notes:			

(a)(i)

M1: Attempts to complete the square on both x and y terms.

Accept $(x \pm 5)^2 + (y \pm 8)^2 = \dots$ or imply this mark for a centre of $(\pm 5, \pm 8)$

Condone $(x \pm 5)^2 \dots (y \pm 8)^2 = \dots$ where the first $\dots$ could be $,$ or even $-$

A1: Correct centre $(5, -8)$.

Accept without brackets. May be written $x = 5, y = -8$

(a)(ii)

A1: 13. The M mark must have been awarded, so it can be scored following a centre of $(\pm 5, \pm 8)$.

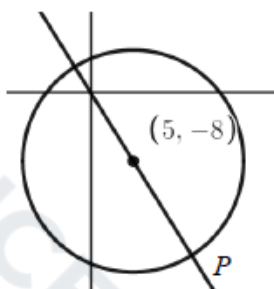
Do not allow for $\sqrt{169}$ or ± 13

(b)

M1: Attempts $\sqrt{5^2 + 8^2} + 13$ for their centre $(5, -8)$ and their radius 13.

Award when this is given as a decimal, e.g. 22.4 for correct centre and radius. Look for $\sqrt{a^2 + b^2} + r$ where centre is $(\pm a, \pm b)$ and radius is r

A1ft: $13 + \sqrt{89}$ Follow through on their $(5, -8)$ and their 13 leading to an exact answer. ISW for example if they write $13 + \sqrt{89} = 22.4$



There are more complicated attempts which could involve finding P by solving $y = -\frac{8}{5}x$ and

$x^2 + y^2 - 10x + 16y = 80$ simultaneously and choosing the coordinate with the greatest modulus. The method is only scored when the distance of the largest coordinate from O is attempted. Such methods are unlikely to result in an exact value but can score 1 mark for the method. Condone slips

FYI. Solving $y = -\frac{8}{5}x$ and $x^2 + y^2 - 10x + 16y = 80 \Rightarrow 89x^2 - 890x - 2000 = 0 \Rightarrow P = (11.89, -19.02)$

Hence $OP = \sqrt{11.89^2 + 19.02^2} (= 22.43)$ scores M1 A0 but $OP = \sqrt{258 + 26\sqrt{89}}$ is M1 A1

Q3.

Question	Scheme	Marks	AOs
(a)	Attempts $(x-2)^2 + (y+5)^2 = \dots$	M1	1.1b
	Centre $(2, -5)$	A1	1.1b
		(2)	
(b)	Sets $k + 2^2 + 5^2 > 0$	M1	2.2a
	$\Rightarrow k > -29$	A1ft	1.1b
		(2)	
(4 marks)			
Notes:			
(a)			
M1: Attempts to complete the square so allow $(x-2)^2 + (y+5)^2 = \dots$			
A1: States the centre is at $(2, -5)$. Also allow written separately $x=2, y=-5$ $(2, -5)$ implies both marks			
(b)			
M1: Deduces that the right hand side of their $(x \pm \dots)^2 + (y \pm \dots)^2 = \dots$ is > 0 or ≥ 0			
A1ft: $k > -29$ Also allow $k \geq -29$ Follow through on their rhs of $(x \pm \dots)^2 + (y \pm \dots)^2 = \dots$			

Q4.

Question	Scheme	Marks	AOs
(a)	C is $(x-r)^2 + (y-r)^2 = r^2$ or $x^2 + y^2 - 2rx - 2ry + r^2 = 0$	B1	2.2a
	$y = 12 - 2x, x^2 + y^2 - 2rx - 2ry + r^2 = 0$ $\Rightarrow x^2 + (12 - 2x)^2 - 2rx - 2r(12 - 2x) + r^2 = 0$ or	M1	1.1b

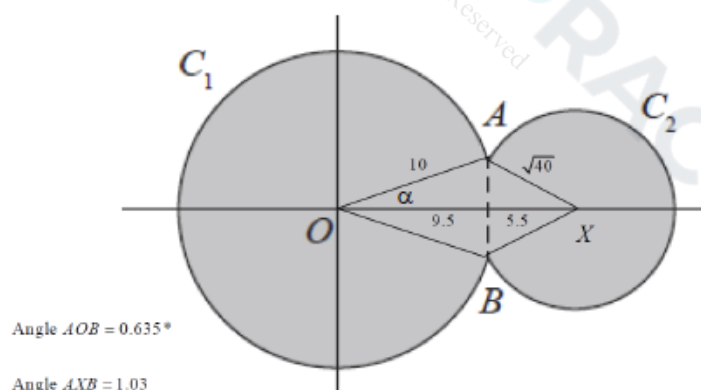
	$y = 12 - 2x, \quad (x-r)^2 + (y-r)^2 = r^2$ $\Rightarrow (x-r)^2 + (12-2x-r)^2 = r^2$		
	$x^2 + 144 - 48x + 4x^2 - 2rx - 24r + 4rx + r^2 = 0$ $\Rightarrow 5x^2 + (2r-48)x + (r^2-24r+144) = 0 \quad *$	A1*	2.1
		(3)	
(b)	$b^2 - 4ac = 0 \Rightarrow (2r-48)^2 - 4 \times 5 \times (r^2-24r+144) = 0$	M1	3.1a
	$r^2 - 18r + 36 = 0 \text{ or any multiple of this equation}$	A1	1.1b
	$\Rightarrow (r-9)^2 - 81 + 36 = 0 \Rightarrow r = \dots$	dM1	1.1b
	$r = 9 \pm 3\sqrt{5}$	A1	1.1b
		(4)	
(7 marks)			

Notes:

- (a)
- B1:** Deduces the correct equation of the circle
- M1:** Attempts to form an equation with terms of the form x^2 , x , r^2 , and xr only using $y = 12 \pm 2x$ and their circle equation which must be of an appropriate form. I.e. includes or implies an x^2 , y^2 , r^2 such as $x^2 + y^2 = r^2$
- If their circle equation starts off as e.g. $(x \pm a)^2 + (y \pm b)^2 = r^2$ then the B mark and the M mark can be awarded when the "a" and "b" are replaced by r or $-r$ as appropriate for their circle equation.
- A1*:** Uses correct and accurate algebra leading to the given solution.
- (b)
- M1:** Attempts to use $b^2 - 4ac \dots 0$ o.e. with $a = 5$, $b = 2r - 48$, $c = r^2 - 24r + 144$ and where ... is "=" or any inequality
- Allow minor slips when copying the a , b and c provided it does not make the work easier and allow their a , b and c if they are similar expressions.
- FYI $(2r-48)^2 - 4 \times 5 \times (r^2-24r+144) = 4r^2 - 192r + 2304 - 20r^2 + 480r - 2880 = -16r^2 + 288r - 576$
- A1:** Correct quadratic equation in r (or inequality). Terms need not be all one side but must be collected.
- E.g. allow $r^2 - 18r = -36$ and allow any multiple of this equation (or inequality).
- dM1:** Correct attempt to solve their 3TQ in r . Dependent upon previous M
- A1:** Careful and accurate work leading to both answers in the required form (must be simplified surds)

Q5.

Question	Scheme	Marks	AOs
(a)	Solves $x^2 + y^2 = 100$ and $(x - 15)^2 + y^2 = 40$ simultaneously to find x or y E.g. $(x - 15)^2 + 100 - x^2 = 40 \Rightarrow x = \dots$	M1	3.1a
	Either $\Rightarrow -30x + 325 = 40 \Rightarrow x = 9.5$ Or $y = \frac{\sqrt{39}}{2} = \text{awrt } \pm 3.12$	A1	1.1b
	Attempts to find the angle AOB in circle C_1 Eg Attempts $\cos \alpha = \frac{9.5}{10}$ to find α then $\times 2$	M1	3.1a
	Angle $AOB = 2 \times \arccos\left(\frac{9.5}{10}\right) = 0.635 \text{ rads (3sf)}$ *	A1*	2.1
		(4)	
(b)	Attempts $10 \times (2\pi - 0.635) = 56.48$	M1	1.1b
	Attempts to find angle AXB or AXO in circle C_2 (see diagram) E.g. $\cos \beta = \frac{15 - 9.5}{\sqrt{40}} \Rightarrow \beta = \dots$ (Note $AXB = 1.03$ rads)	M1	3.1a
	Attempts $10 \times (2\pi - 0.635) + \sqrt{40} \times (2\pi - 2\beta)$	dM1	2.1
	$= 89.7$	A1	1.1b
		(4)	
(8 marks)			
Notes:			



(a)

M1: For the key step in an attempt to find either coordinate for where the two circles meet.

Look for an attempt to set up an equation in a single variable leading to a value for x or y .

A1: $x = 9.5$ (or $y = \frac{\sqrt{39}}{2} = \text{awrt } \pm 3.12$)

M1: Uses the radius of the circle and correct trigonometry in an attempt to find angle AOB in circle C_1

E.g. Attempts $\cos \alpha = \frac{9.5}{10}$ to find α then $\times 2$

Alternatives include $\tan \alpha = \frac{\sqrt{100 - 9.5^2}}{9.5} = (0.3286...)$ to find α then $\times 2$

$$\text{And } \cos AOB = \frac{10^2 + 10^2 - (\sqrt{39})^2}{2 \times 10 \times 10} = \frac{161}{200}$$

A1*: Correct and careful work in proceeding to the given answer. Condone an answer with greater accuracy.

Condone a solution where the intermediate value has been truncated, provided the trig equation is correct.

E.g. $\sin \alpha = \frac{\sqrt{39}}{20} \Rightarrow \alpha = 0.317 \Rightarrow AOB = 2\alpha = 0.635$

Condone a solution written down from awrt 36.4° (without the need to shown any calculation.)

E

(b)

M1: Attempts to use the formula $s = r\theta$ with $r = 10$ and $\theta = 2\pi - 0.635$

The formula may be embedded. You may see $\frac{2\pi \times 10 + 2\pi \sqrt{40} - 10 \times 0.635}{2\pi}$ which is fine for this M1

M1: Attempts to use a correct method in order to find angle AXB or AXO in circle C_2

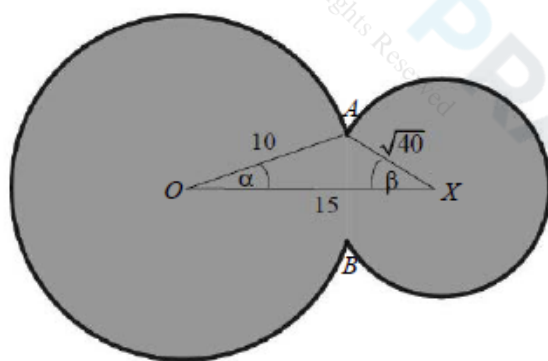
Amongst many other methods are $\tan \beta = \frac{3.12}{15 - 9.5}$ and $\cos AXB = \frac{40 + 40 - (\sqrt{39})^2}{2 \times \sqrt{40} \times \sqrt{40}} = \frac{41}{80}$

Note that many candidates believe this to be 0.635. This scores M0 dM0 A0

dM1: A full and complete attempt to find the perimeter of the region.

It is dependent upon having scored both M's.

A1: awrt 89.7



(a)

M1: For the key step in attempting to find all lengths in triangle OAX , condoning slips

A1: All three lengths correct

M1: Attempts cosine rule to find α then $\times 2$

A1*: Correct and careful work in proceeding to the given answer

Q6.

Question	Scheme	Marks	AOs
(a)	Deduces that gradient of PA is $-\frac{1}{2}$	M1	2.2a
	Finding the equation of a line with gradient " $-\frac{1}{2}$ " and point $(7,5)$ $y - 5 = -\frac{1}{2}(x - 7)$	M1	1.1b
	Completes proof $2y + x = 17$ *	A1*	1.1b
		(3)	
(b)	Solves $2y + x = 17$ and $y = 2x + 1$ simultaneously	M1	2.1
	$P = (3, 7)$	A1	1.1b
	Length $PA = \sqrt{(3-7)^2 + (7-5)^2} = (\sqrt{20})$	M1	1.1b
	Equation of C is $(x-7)^2 + (y-5)^2 = 20$	A1	1.1b
		(4)	
(c)	Attempts to find where $y = 2x + k$ meets C using $\overrightarrow{OA} + \overrightarrow{PA}$	M1	3.1a
	Substitutes their $(11, 3)$ in $y = 2x + k$ to find k	M1	2.1
	$k = -19$	A1	1.1b
		(3)	
(10 marks)			

(c)	Attempts to find where $y = 2x + k$ meets C via simultaneous equations proceeding to a 3TQ in x (or y) FYI $5x^2 + (4k - 34)x + k^2 - 10k + 54 = 0$	M1	3.1a
	Uses $b^2 - 4ac = 0$ oe and proceeds to $k = \dots$	M1	2.1
	$k = -19$	A1	1.1b
		(3)	

Notes:

(a)

M1: Uses the idea of perpendicular gradients to deduce that gradient of PA is $-\frac{1}{2}$. Condone $-\frac{1}{2}x$ if followed by correct work. You may well see the perpendicular line set up as $y = -\frac{1}{2}x + c$ which scored this mark

M1: Award for the method of finding the equation of a line with a changed gradient and the point $(7, 5)$

So sight of $y - 5 = \frac{1}{2}(x - 7)$ would score this mark

If the form $y = mx + c$ is used expect the candidates to proceed as far as $c = \dots$ to score this mark.

A1*: Completes proof with no errors or omissions $2y + x = 17$

(b)

M1: Awarded for an attempt at the key step of finding the coordinates of point P , ie for an attempt at solving $2y + x = 17$ and $y = 2x + 1$ simultaneously. Allow any methods (including use of a calculator) but it must be a valid attempt to find both coordinates. Do not allow where they start $17 - x = 2x + 1$ as they have set $2y = y$ but condone bracketing errors, eg $2 \times 2x + 1 + x = 17$

A1: $P = (3, 7)$

M1: Uses Pythagoras' Theorem to find the radius or radius ² using their $P = (3, 7)$ and $(7, 5)$. There must be an attempt to find the difference between the coordinates in the use of Pythagoras

A1: $(x-7)^2 + (y-5)^2 = 20$. Do not accept $(x-7)^2 + (y-5)^2 = (\sqrt{20})^2$

(c)

M1: Attempts to find where $y = 2x + k$ meets C .

Awarded for using $\overrightarrow{OA} + \overrightarrow{PA}$. $(11, 3)$ or one correct coordinate of $(11, 3)$ is evidence of this award.

M1: For a full method leading to k . Scored for either substituting their $(11, 3)$ in $y = 2x + k$

or, in the alternative, for solving their $(4k-34)^2 - 4 \times 5 \times (k^2 - 10k + 54) = 0 \Rightarrow k = \dots$ Allow use of a calculator here to find roots. Award if you see use of correct formula but it would be implied by $\pm$ correct roots

A1: $k = -19$ only

Alternative I

M1: For solving $y = 2x + k$ with their $(x-7)^2 + (y-5)^2 = 20$ and creating a quadratic eqn of the form $ax^2 + bx + c = 0$ where both b and c are dependent upon k . The terms in x^2 and x must be collected together or implied to have been collected by their correct use in " $b^2 - 4ac$ "

FYI the correct quadratic is $5x^2 + (4k-34)x + k^2 - 10k + 54 = 0$

M1: For using the discriminant condition $b^2 - 4ac = 0$ to find k . It is not dependent upon the previous M and may be awarded from only one term in k .

$(4k-34)^2 - 4 \times 5 \times (k^2 - 10k + 54) = 0 \Rightarrow k = \dots$ Allow use of a calculator here to find roots.

Award if you see use of correct formula but it would be implied by $\pm$ correct roots

A1: $k = -19$ only

Alternative II

M1: For solving $2y + x = 17$ with their $(x-7)^2 + (y-5)^2 = 20$, creating a 3TQ and solving.

M1: For substituting their $(11, 3)$ into $y = 2x + k$ and finding k

A1: $k = -19$ only

.....
Other method are possible using trigonometry.